DETAILED DESCRIPTION OF THE INVENTION

Manufacturing and Calibration Architecture

In various embodiments, the penile anatomical geometry profiling device is manufactured according to predefined mechanical and electronic tolerance specifications to preserve distance measurement accuracy and motion tracking consistency. The housing structure may be formed using precision-molded polymers, composite materials, machined metals, or hybrid assemblies designed to maintain fixed spatial relationships between sensor reference planes and external contact surfaces.

During manufacturing, individual time-of-flight (ToF) sensors may undergo factory calibration to establish baseline ranging offsets, signal timing parameters, and reflectivity compensation values. Calibration constants may be stored in non-volatile memory within the device and referenced during operation to correct systematic measurement bias. In embodiments utilizing multiple ToF sensors or multi-zone depth arrays, inter-sensor alignment calibration may be performed to ensure geometric coherence across measurement zones.

Optical motion sensing components may similarly be calibrated to establish displacement scaling factors, surface reflectivity response thresholds, and drift compensation parameters. In some implementations, calibration may include controlled motion testing across known linear distances to generate correction coefficients applied during operational displacement tracking.

The device may further include self-calibration routines executed during startup or periodically during use. Such routines may verify sensor responsiveness, confirm baseline distance values relative to internal reference surfaces, and detect abnormal deviations indicating hardware misalignment or environmental interference.

In certain embodiments, calibration artifacts such as reference blocks, controlled-distance fixtures, or embedded reference reflectors may be used during manufacturing or service procedures to validate ranging accuracy. Software-based calibration tools may update correction coefficients via secure firmware updates while preserving cryptographic integrity verification mechanisms.

Manufacturing and calibration architectures described herein are not limited to any specific fabrication method, material selection, tolerance range, or calibration procedure, provided that the resulting system maintains sufficient spatial accuracy to generate a reliable user-specific penile anatomical geometry profile derived from synchronized distance and motion measurements.